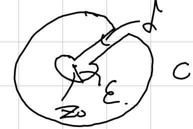
「Thm. Cauchy Integral Formular.

suppose f is Holomorphic on some open set containing a closure of disk D.
If C is a curve with positive orientation of ∂D, then for any $z_0 \in D$,

$$f(z_0) = \frac{1}{2\pi i} \cdot \int_C \frac{f(z)}{z - z_0}\, dz$$

Proof. Let $z_0 \in D$ be fixed, and consider a curve $\Gamma_{\delta,\varepsilon}$ as



Since $\dfrac{f(z)}{z - z_0}$ is Holomorphic on $D \setminus \{z_0\}$, therefore

$\displaystyle\int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = 0$. Taking $\delta \to 0$, then we obtain

$$\lim_{\delta \to 0} \int_{\Gamma_{\delta,\varepsilon}} \frac{f(z)}{z - z_0}\, dz = \int_{C_\varepsilon} \frac{f(z)}{z - z_0}\, dz + \int_C \frac{f(z)}{z - z_0}\, dz = 0.$$

Meanwhile, $\dfrac{f(z)}{z - z_0} = \dfrac{f(z) - f(z_0)}{z - z_0} + \dfrac{f(z_0)}{z - z_0}$ and since f is Holomorphic,
the first term is bounded, therefore $\displaystyle\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z) - f(z_0)}{z - z_0}\, dz = 0$.

Now,
$$\lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{f(z_0)}{z - z_0}\, dz = f(z_0) \cdot \lim_{\varepsilon \to 0} \int_{C_\varepsilon} \frac{1}{z - z_0}\, dz$$
$$= f(z_0) \cdot \lim_{\varepsilon \to 0} \left(- \int_0^{2\pi} \frac{1}{\varepsilon \cdot e^{it}} \cdot \varepsilon \cdot i \cdot e^{it}\, dt \right)$$
$$= - f(z_0) \cdot 2\pi i.$$

Consequently, $\displaystyle\int_C \frac{f(z)}{z - z_0}\, dz = 2\pi i\, f(z_0)$.

「Corollary. $\displaystyle\int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = 2\pi i \cdot \frac{f^{(n)}(z_0)}{n!}$.

Proof. Using induction for n. For $n=0$, clear. If $f^{(n-1)}(z_0) = \dfrac{(n-1)!}{2\pi} \cdot \displaystyle\int_C \frac{f(z)}{(z - z_0)^n}\, dz$, then

$$\frac{f^{(n-1)}(z_0 + h) - f^{(n-1)}(z_0)}{h} = \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \left[\frac{1}{(z - z_0 - h)^n} - \frac{1}{(z - z_0)^n} \right] dz$$
$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{h} \cdot \left[\frac{h}{(z - z_0 - h)(z - z_0)} \cdot \left(\sum_{k=0}^{n-1} \frac{1}{(z - z_0 - h)^{n-1-k}} \cdot \frac{1}{(z - z_0)^k} \right) \right] dz$$
$$\to \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot \frac{1}{(z - z_0)^2} \cdot \sum_{k=0}^{n-1} \frac{1}{(z - z_0)^{n-1}}\, dz \quad \text{as } h \to 0$$
$$= \frac{(n-1)!}{2\pi i} \cdot \int_C f(z) \cdot n \cdot \frac{1}{(z - z_0)^{n+1}}\, dz$$
$$= \frac{n!}{2\pi i} \cdot \int_C \frac{f(z)}{(z - z_0)^{n+1}}\, dz = f^{(n)}(z_0).$$

「Thm. Cauchy inequality.

$|f^{(n)}(z_0)| \le \dfrac{n!}{R} \cdot \sup_C |f|$ if f is Holomorphic in an open containing $\overline{B_R(z_0)}$, $C = \partial B_R(z_0)$.

「Thm. Suppose f is Holomorphic on an open Ω Containing a closed disk D, having center z_0,

$$f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n \qquad (z \in D)$$

Proof. $$f(z) = \frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{\mu - z} \, d\mu \qquad \left(\frac{1}{\mu - z} = \frac{1}{\mu - z_0 - (z - z_0)} = \frac{1}{\mu - z_0} \cdot \frac{1}{1 - \left(\frac{z - z_0}{\mu - z_0} \right)} \right)$$

$$= \frac{1}{2\pi i} \cdot \int_C f(\mu) \cdot \sum_{n=0}^{\infty} \frac{1}{(\mu - z_0)^{n+1}} \cdot (z - z_0)^n \, d\mu$$

$$= \sum_{n=0}^{\infty} \left(\frac{1}{2\pi i} \cdot \int_C \frac{f(\mu)}{(\mu - z_0)^{n+1}} \, d\mu \right) \cdot (z - z_0)^n$$

$$= \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} \cdot (z - z_0)^n.$$

「Corollary, Liouville's Theorem.

If $f : \mathbb{C} \to \mathbb{C}$ is entire and bounded, then it is Constant.

Proof. Given $z_0 \in \mathbb{C}$ and $R > 0$, $|f'(z_0)| \le \frac{1}{R} \cdot \sup_{C_R} |f| \to 0$ as $R \to \infty$. Therefore, $f' = 0$

Thus f is Constant.

「Fundamental Theorem of Algebra.

Non-constant polynomial $P(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_0 \in \mathbb{C}[z]$ has a root in $\mathbb{C}$.

Proof. Suppose that $P(z)$ has no root in $\mathbb{C}$. Then, $1/P(z)$ is well-defined on $\mathbb{C}$, entire.

observe that $$\frac{P(z)}{z^n} = a_n + \left(\frac{a_{n-1}}{z} + \cdots + \frac{a_0}{z^n} \right) \qquad (z \ne 0).$$

The terms $a_{n-k} \cdot z^{-k}$ $(k = 1, \cdots, n)$ Converges to zero as $|z| \to \infty$, thus for some $R > 0$,

$$|P(z)| \ge 2|a_n| \cdot z^n \qquad (|z| > R)$$
$$\ge 2|a_n| \cdot R^n$$

Therefore it is bounded below in $|z| > R$, and clearly in $|z| \le R$.

Now, $\frac{1}{P}$ is bounded, thus Constant. Contradiction.